

Juan Nathaniel

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Education

Columbia University

PHD, EARTH AND ENVIRONMENTAL ENGINEERING

• Advisor: Professor Pierre Gentine

New York

Expected 2026

Columbia University

MS, EARTH AND ENVIRONMENTAL ENGINEERING

New York

2021 - 2022

National University of Singapore

BS, ENVIRONMENTAL SCIENCE

Singapore

2015 - 2019

Professional Experience

2023, 2026 **Research Intern**, IBM TJ Watson Research Center

2019 - 2021 **Research Engineer**, Living Analytics Research Center

Publications

* denotes co-first authorship

PUBLISHED

- [1] **Nathaniel** & Gentine (2026). *Generative emulation of chaotic dynamics with coherent prior*. *Computer Methods in Applied Mechanics and Engineering (CMAME)*
- [2] **Nathaniel***, Roesch*, et al (2025). *Deep Koopman operators for causal discovery*. *Communications Physics*
- [3] Herdeanu*, **Nathaniel***, Roesch*, et al (2025). *CausalDynamics: A large-scale benchmark for structural discovery of dynamical causal models*. *Advances in Neural Information Processing Systems 38 (NeurIPS)*
- [4] **Nathaniel**, et al (2024). *Chaosbench: A multi-channel, physics-based benchmark for subseasonal-to-seasonal climate prediction*. *Advances in Neural Information Processing Systems 37 (NeurIPS; Oral Top 0.6%)*
- [5] Qu*, **Nathaniel***, et al (2024). *Deep generative data assimilation in multimodal setting*. *Computer Vision and Pattern Recognition Workshop (CVPRW EarthVision; Best Student Paper)*
- [6] **Nathaniel**, et al (2024). *Inferring failure risk of on-site wastewater systems from physical and social factors*. *npj Clean Water*
- [7] Kim*, **Nathaniel***, et al (2024). *Spatiotemporal upscaling of sparse air-sea pCO₂ data via physics-informed transfer learning*. *Scientific Data*
- [8] **Nathaniel**, et al (2023). *MetaFlux: Meta-learning global carbon fluxes from sparse spatiotemporal observations*. *Scientific Data*

IN PREP / IN REVIEW

- [9] **Nathaniel**, et al (2025). *Multiscale stochastic parameterization with deep Mori-Zwanzig formalism*.
- [10] **Nathaniel**, et al (2026). *Koopman early warning signals for bifurcation and rate-induced tipping*.

[11] **Nathaniel**, et al (2026). *On the sufficiency of probabilistic modeling for causal inference in dynamical systems*.

Awards, Fellowships, & Grants _____

2024 - 2025	Columbia-Dream Sports AI PhD Fellowship , Columbia University	\$ 100,000
Summer 2024	Best Student Paper Award , CVPR Workshop on EarthVision	
2015 - 2019	ASEAN Undergraduate Scholarship , National University of Singapore	\$ 200,000

Invited Talks _____

Spring 2026	Operator Learning for Climate Tipping Point Detection , UK ARIA
Fall 2025	Deep Learning Koopman Causality , Potsdam Institute for Climate Impact Research
Fall 2025	Operator Learning for Climate Modeling Across Scales , University of Utah
Spring 2025	Subseasonal-to-Seasonal Climate Predictability , Stanford University
Fall 2024	Climate Predictability Across Timescales: A Probabilistic Machine Learning Perspective , UCLA
Fall 2023	Understanding and Modelling the Earth System with Machine Learning , German Aerospace Center
Spring 2023	Meta-learning for Environmental Sciences , University of Reading

Teaching Experience _____

Fall 2025	EAAE 2100 , A Better Planet by Design
Spring 2025	Hackathon: Learning the Earth with Artificial Intelligence and Physics , Organizer
Fall 2024	EAAE 4000 , Machine Learning for Environmental Engineering and Science
Summer 2024	Summer Bootcamp: Learning the Earth with Artificial Intelligence and Physics , Bootcamp Instructor in Numerical Climate Modeling

Service _____

2026	Reviewer , Communications Earth & Environment
2026	Reviewer , International Journal of Bifurcation and Chaos (IJBC)
2025, 2026	Reviewer , International Conference on Learning Representations (ICLR)
2024, 2025, 2026	Reviewer , Advances in Neural Information Processing Systems (NeurIPS)
2025	Reviewer , International Conference on Machine Learning (ICML)
2024	Reviewer , Journal of Advances in Modeling Earth Systems (JAMES)